

Image Enhancement:

Filtering in the Frequency Domain

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INTRODUCTION

- ▶ Image enhancement is basically improving the interpretability or perception of information in images for human viewers and providing 'better' input for other automated image processing techniques.
- ▶ The principal objective of image enhancement is to modify attributes of an image to make it more suitable for a given task and a specific observer.
- ▶ During this process, one or more attributes of the image are modified.

IMAGE ENHANCEMENT

- ▶ There exist many techniques that can enhance a digital image without spoiling it.
- ▶ The enhancement methods can broadly be divided in to the following two categories:
 - Spatial Domain Methods
 - Frequency Domain Methods

IMAGE ENHANCEMENT

- ▶ Unfortunately, there is no general theory for determining what is 'good' image enhancement when it comes to human perception. If it looks good, it is good.
- ▶ However, when image enhancement techniques are used as pre-processing tools for other image processing techniques, then quantitative measures can determine which techniques are most appropriate.

SPATIAL DOMAIN ENHANCEMENT

Spatial Domain Methods

- ▶ In spatial domain techniques , we directly deal with the image pixels.
- ▶ The pixel values are manipulated to achieve desired enhancement.
- ▶ The value of a pixel with coordinates $(x; y)$ in the enhanced image 'F' is the result of performing some operation on the pixels in the neighborhood of $(x; y)$ in the input image 'f'.

FREQUENCY DOMAIN ENHANCEMENT

- ▶ In frequency domain methods, the image is first transferred into frequency domain.
- ▶ It means that, the Fourier Transform of the image is computed first.
- ▶ All the enhancement operations are performed on the Fourier transform of the image and then the Inverse Fourier transform is performed to get the resultant image.

FREQUENCY DOMAIN ENHANCEMENT

- ▶ These enhancement operations are performed in order to modify the image brightness, contrast or the distribution of the grey levels.
- ▶ As a consequence the pixel value (intensities) of the output image will be modified according to the transformation function applied on the input values.
- ▶ Image enhancement is applied in every field for example medical image analysis, analysis of images from satellites etc.

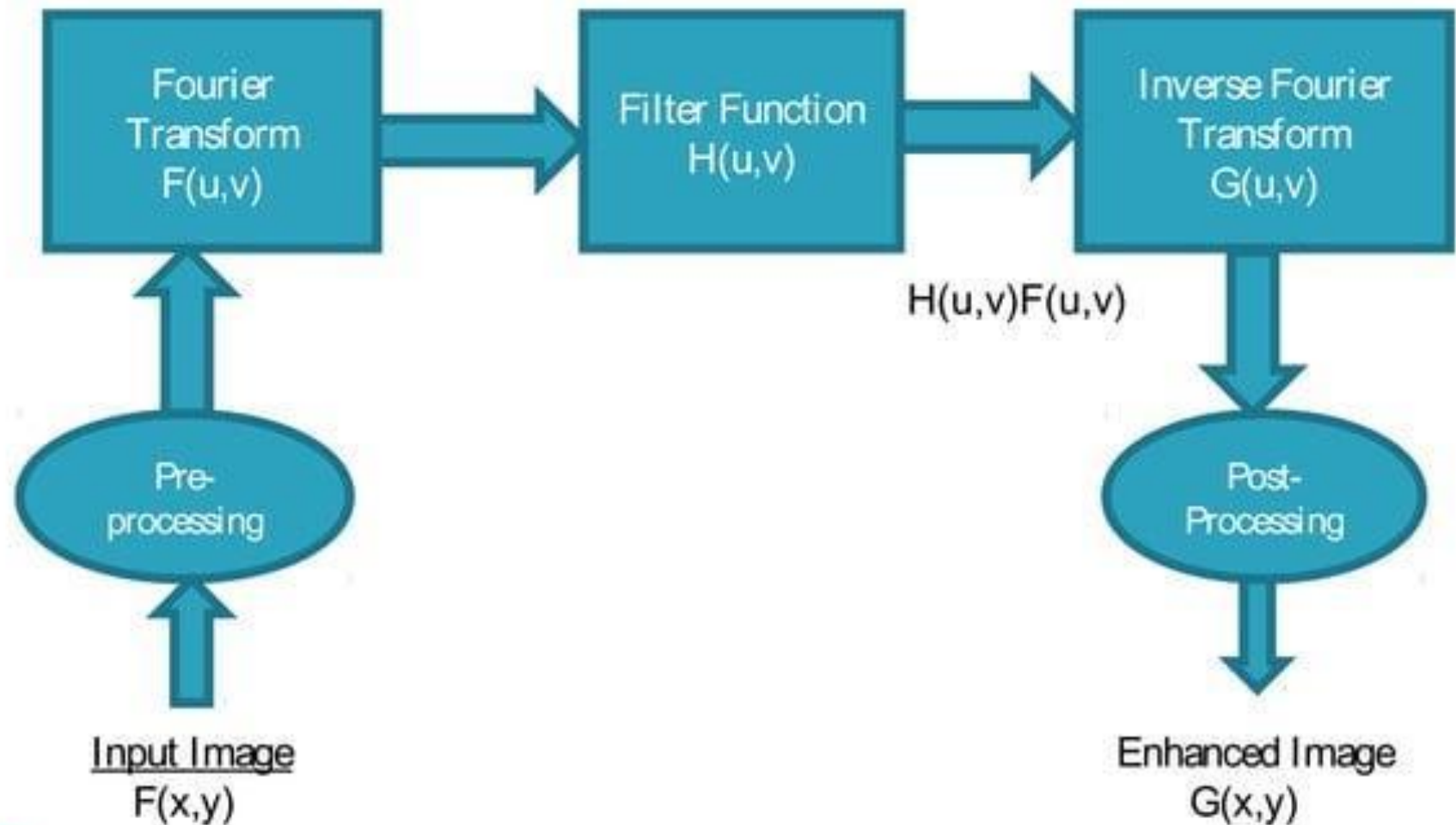
FREQUENCY DOMAIN ENHANCEMENT

- ▶ Image enhancement simply means, transforming an image F into image G using T . (Where T is the transformation function.
- ▶ The values of pixels in images F and G are denoted by r and s , respectively. As said, the pixel values r and s are related by the expression

$$s = T(r)$$

- Where T is a transformation that maps a pixel value r into a pixel value s .

FREQUENCY DOMAIN ENHANCEMENT



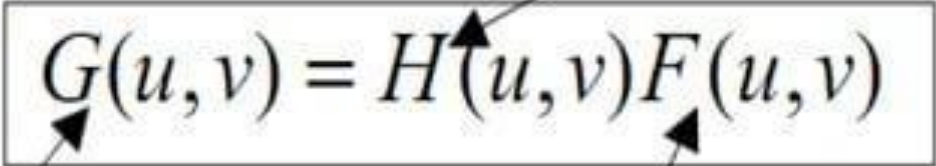
Frequency domain filtering operations

To filter an image in the frequency domain:

1. Compute $F(u,v)$ the DFT of the image
2. Multiply $F(u,v)$ by a filter function $H(u,v)$
3. Compute the inverse DFT of the result

FREQUENCY DOMAIN ENHANCEMENT

- ▶ We can therefore directly design a transfer function and implement the enhancement in the frequency domain as follows.



The diagram shows the equation $G(u, v) = H(u, v)F(u, v)$ enclosed in a rectangular box. Three arrows point from external labels to terms in the equation: one from 'Enhanced Image' to $G(u, v)$, one from 'Transfer function' to $H(u, v)$, and one from 'Given Image' to $F(u, v)$.

$$G(u, v) = H(u, v)F(u, v)$$

Enhanced Image

Transfer function

Given Image

FILTERING

- ▶ The concept of filtering is easier to visualize in the frequency domain.
- ▶ Therefore, enhancement of image can be done in the frequency domain, based on its DFT.

FILTERING

- ▶ Filtering can be divided in two categories namely
- ▶ Low pass filtering
- ▶ High Pass filtering

FILTERING

- ▶ Edges and sharp transitions in gray values in an image contribute significantly to high-frequency content of its Fourier transform.
- ▶ Regions of relatively uniform gray values in an image contribute to low-frequency content of its Fourier transform.

LOW PASS FILTERING

- ▶ Hence, an image can be smoothed in the Frequency domain by attenuating the high-frequency content of its Fourier transform.
- ▶ This would be a low pass filter.
- ▶ For simplicity, we will consider only those filters that are real and symmetric.

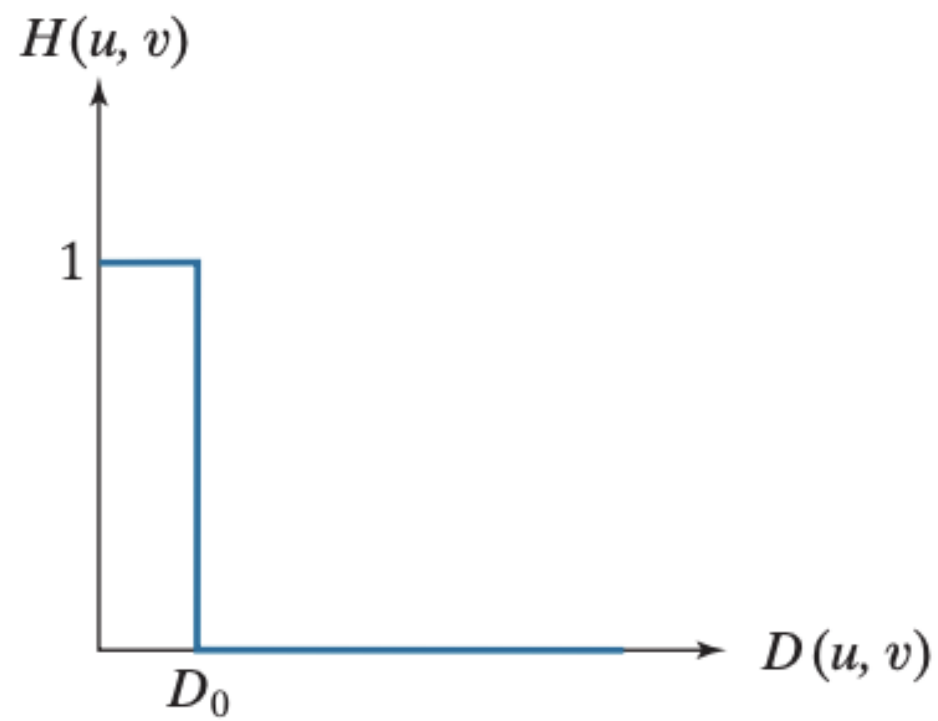
IDEAL LOWPASS FILTERS

A 2-D lowpass filter that passes without attenuation all frequencies within a circle of radius from the origin, and “cuts off” all frequencies outside this, circle is called an *ideal lowpass filter* (ILPF); it is specified by the transfer function

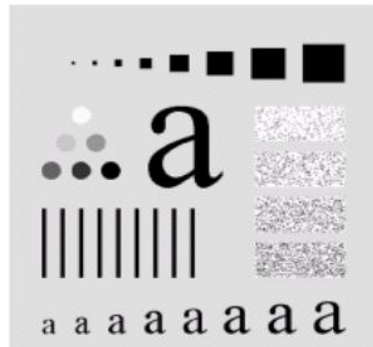
$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \leq D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases} \quad (4-111)$$

where D_0 is a positive constant, and $D(u,v)$ is the distance between a point (u,v) in the frequency domain and the center of the $P \times Q$ frequency rectangle; that is,

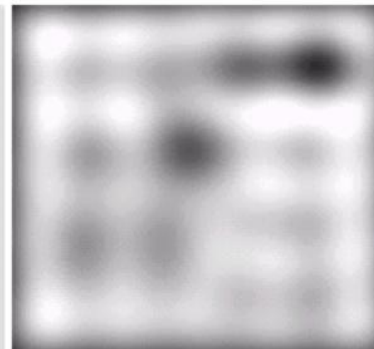
$$D(u,v) = \left[(u - P/2)^2 + (v - Q/2)^2 \right]^{1/2} \quad (4-112)$$



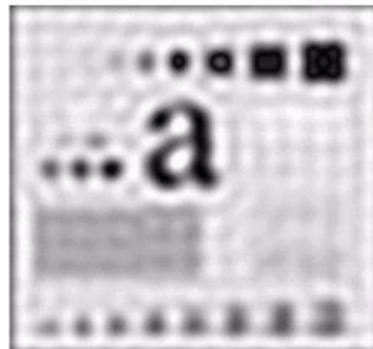
Original
image



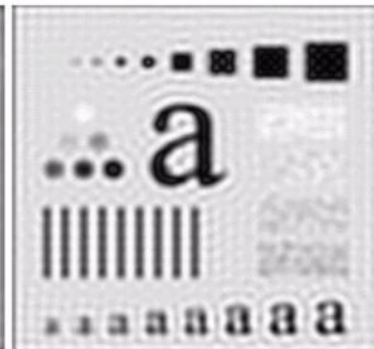
Result of filtering
with ideal low pass
filter of radius 5



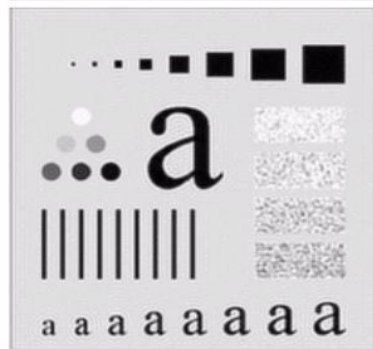
Result of filtering
with ideal low pass
filter of radius 15



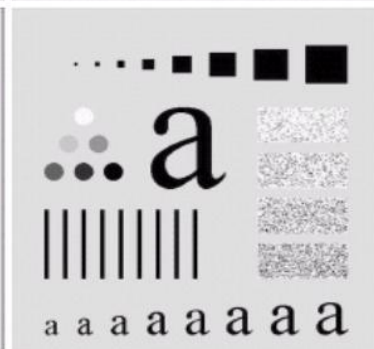
Result of filtering
with ideal low pass
filter of radius 30



Result of filtering
with ideal low pass
filter of radius 80



Result of filtering
with ideal low pass
filter of radius 230



LOW PASS FILTERING

- ❖ The cutoff frequency of the ideal LPF determines the amount of frequency components passed by the filter.
- ❖ Smaller the value of r_0 , more the number of image components eliminated by the filter.
- ❖ In general, the value of r_0 is chosen such that most components of interest are passed through, while most components not of interest are eliminated.

$$*D_0=r_0$$

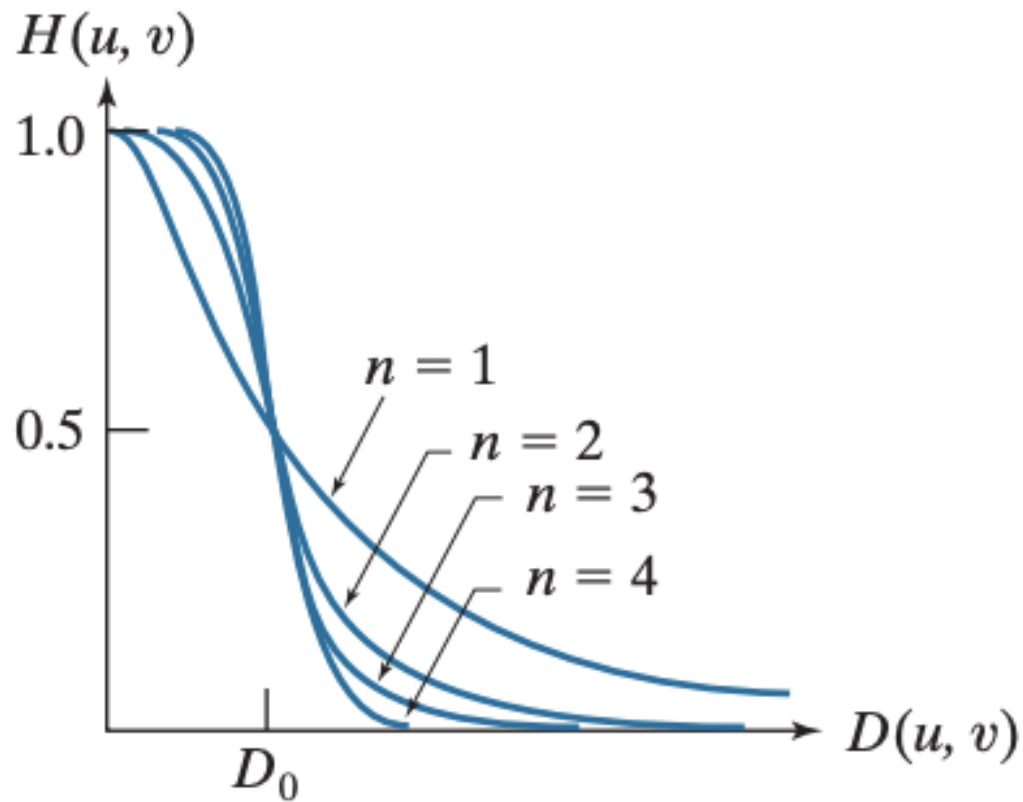
BUTTERWORTH LOW PASS FILTERING

- ▶ A two-dimensional Butterworth low pass filter has transfer function:

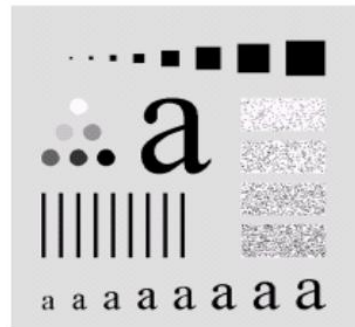
$$H(u,v) = \frac{1}{1 + [D(u,v)/D_0]^{2n}}$$

n : filter order, D_0 : Cut off frequency

Butterworth Low Pass Filter



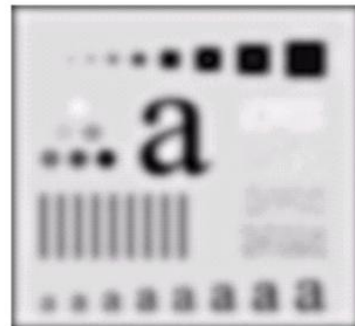
Original
image



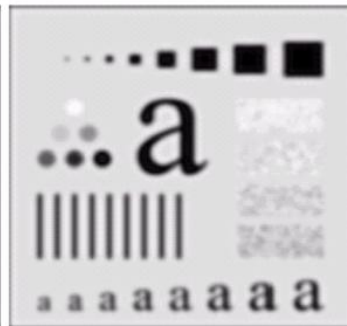
Result of filtering
with Butterworth filter
of order 2 and cutoff
radius 5



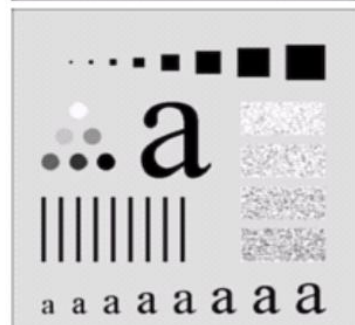
Result of filtering with
Butterworth filter of
order 2 and cutoff
radius 15



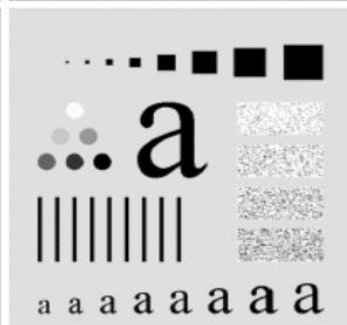
Result of filtering
with Butterworth filter
of order 2 and
cutoff radius 30



Result of filtering with
Butterworth filter of
order 2 and cutoff
radius 80



Result of filtering
with Butterworth filter
of order 2 and
cutoff radius 230



BUTTERWORTH LOW PASS FILTERING

- ▶ Frequency response does not have a sharp transition as in the ideal LPF.
- ▶ This is more appropriate for image smoothing than the ideal LPF, since this not introduce ringing.

GAUSSIAN LOW PASS FILTERING

- ▶ The form of a Gaussian low pass filter in two-dimensions is given by

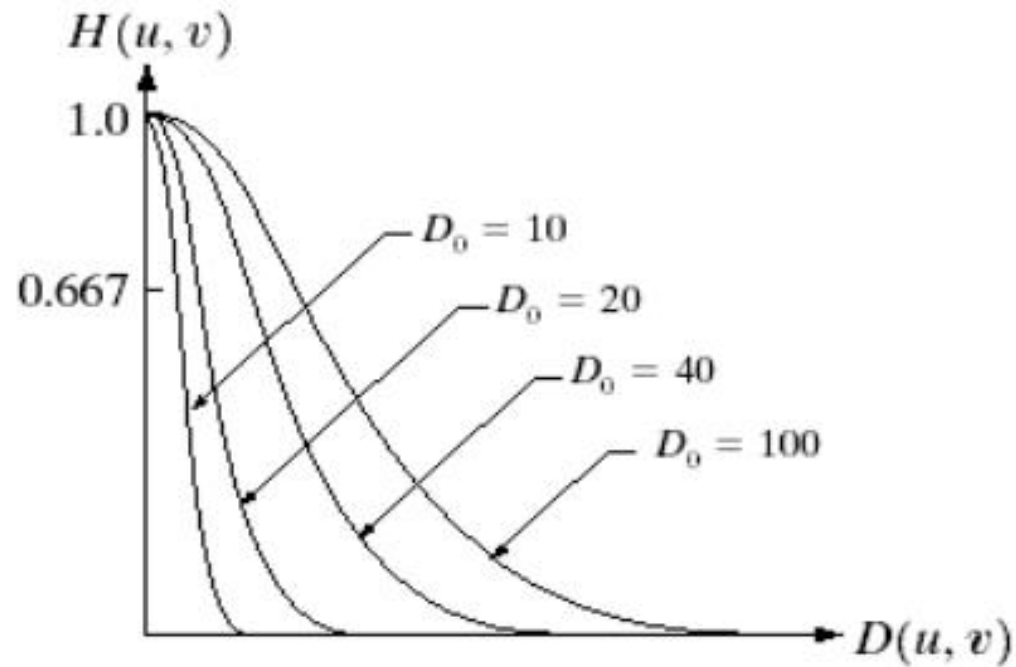
$$H(u,v) = e^{-D^2(u,v)/2D_0^2} \quad D(u,v) = \sqrt{u^2 + v^2}$$

□ Where D is the distance from origin in frequency plane

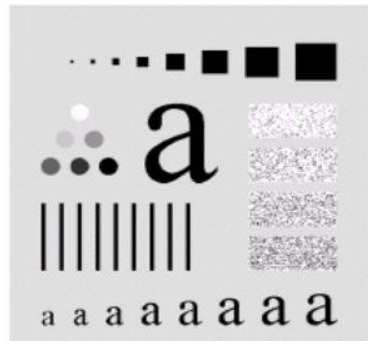
❖ The parameter σ measures the dispersion of the Gaussian curve. Larger the value of σ , larger the cutoff frequency and milder the filtering.

$$\sigma = D_0$$

Gaussian Low Pass



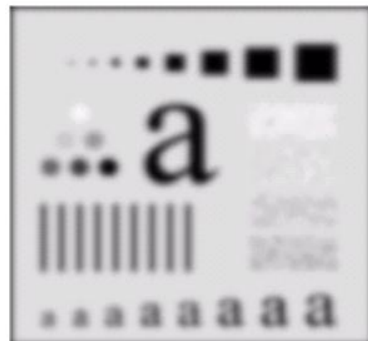
Original
image



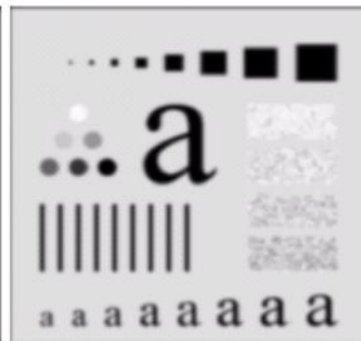
Result of filtering
with Gaussian filter
with cutoff radius 5



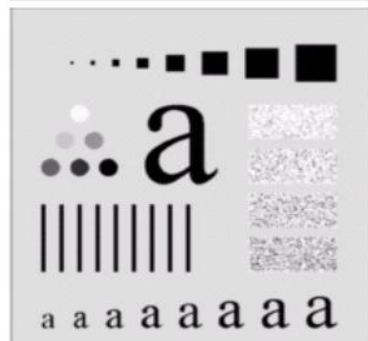
Result of filtering
with Gaussian
filter with cutoff
radius 15



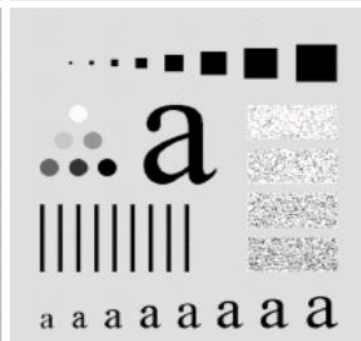
Result of filtering
with Gaussian filter
with cutoff radius 30



Result of filtering
with Gaussian
filter with cutoff
radius 85



Result of filtering
with Gaussian filter
with cutoff radius
230



HIGH PASS FILTERING

- Edges and fine detail characterized by sharp transitions in image intensity
- Such transitions contribute significantly to high frequency components of Fourier transform
- Intuitively, attenuating certain low frequency components and preserving high frequency components result in sharpening

High pass filters – only pass the high frequencies, drop the low ones

High pass frequencies are precisely the reverse of low pass filters, so:

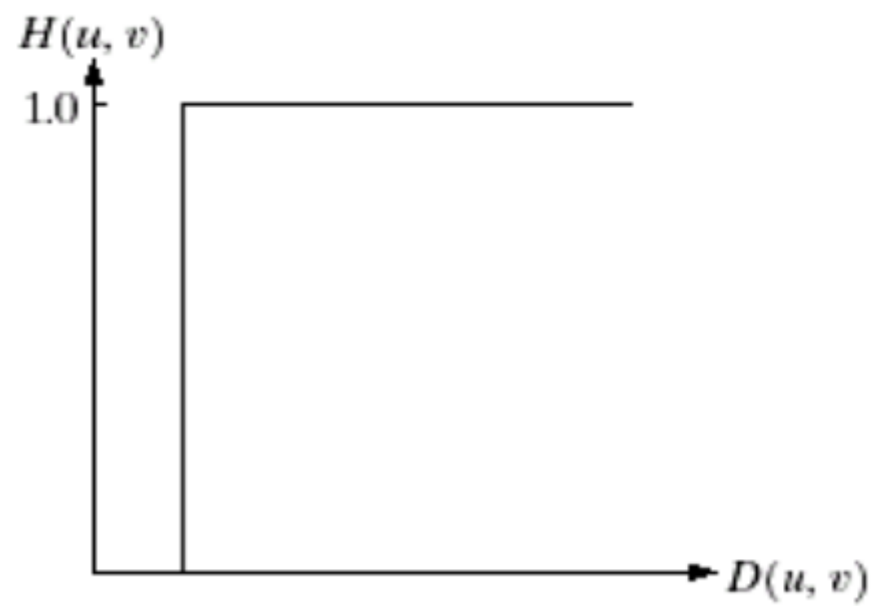
$$H_{hp}(u, v) = 1 - H_{lp}(u, v)$$

Ideal High Pass Filter

The ideal high pass filter is given as:

$$H(u, v) = \begin{cases} 0 & \text{if } D(u, v) \leq D_0 \\ 1 & \text{if } D(u, v) > D_0 \end{cases}$$

where D_0 is the cut off distance as before

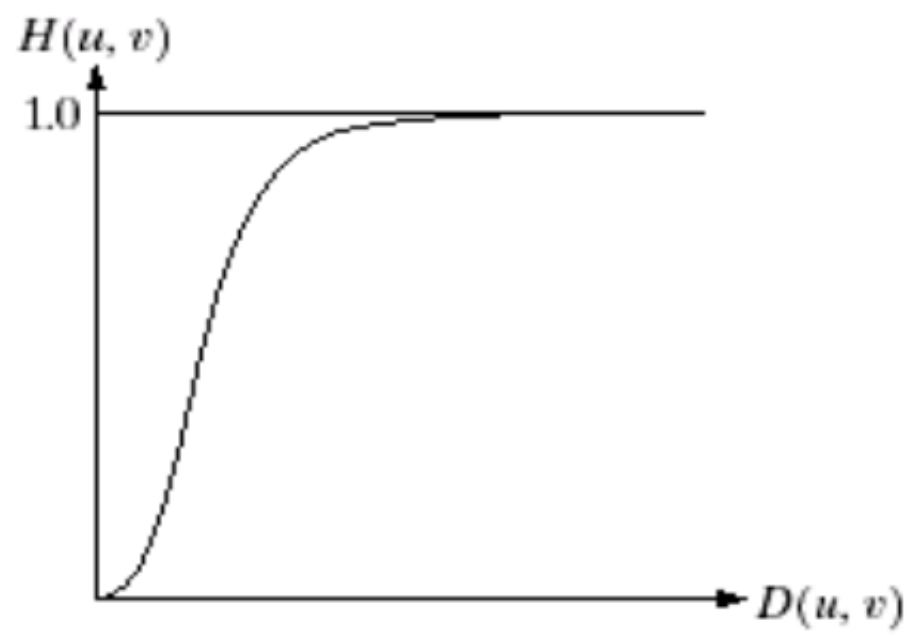


BUTTERWORTH HIGH PASS FILTERING

The Butterworth high pass filter is given as:

$$H(u, v) = \frac{1}{1 + [D_0 / D(u, v)]^{2n}}$$

where n is the order and D_0 is the cut off distance as before

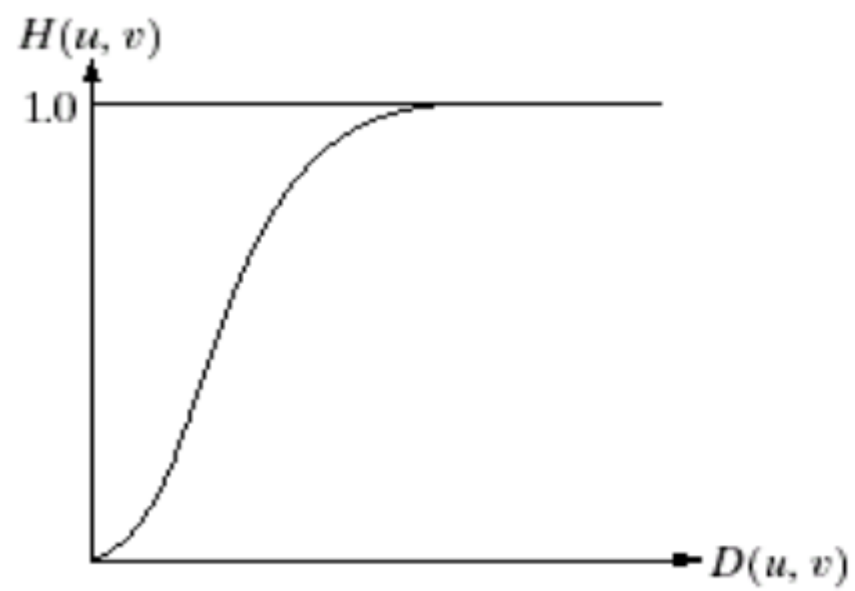


Gaussian High Pass Filter

The Gaussian high pass filter is given as:

$$H(u, v) = 1 - e^{-D^2(u, v) / 2D_0^2}$$

where D_0 is the cut off distance as before



CONCLUSION

- ▶ Image enhancement algorithms offer a wide variety of approaches for modifying images to achieve visually acceptable images.
- ▶ The choice of such techniques is a function of the specific task, image content, observer characteristics, and viewing conditions.